

IEOR 4101: Probability, Statistics and Simulation

Lecture 12: Interval Estimation

Confidence Interval

- Suppose $X_1, X_2, \dots, X_n$ is an i.i.d. sample from a population with an unknown parameter, say μ
- We have discussed how to find a *point estimator* for μ (i.e., a single number that summarizes our best guess)
- However, a point estimate cannot tell us the reliability of our best guess

Confidence Interval

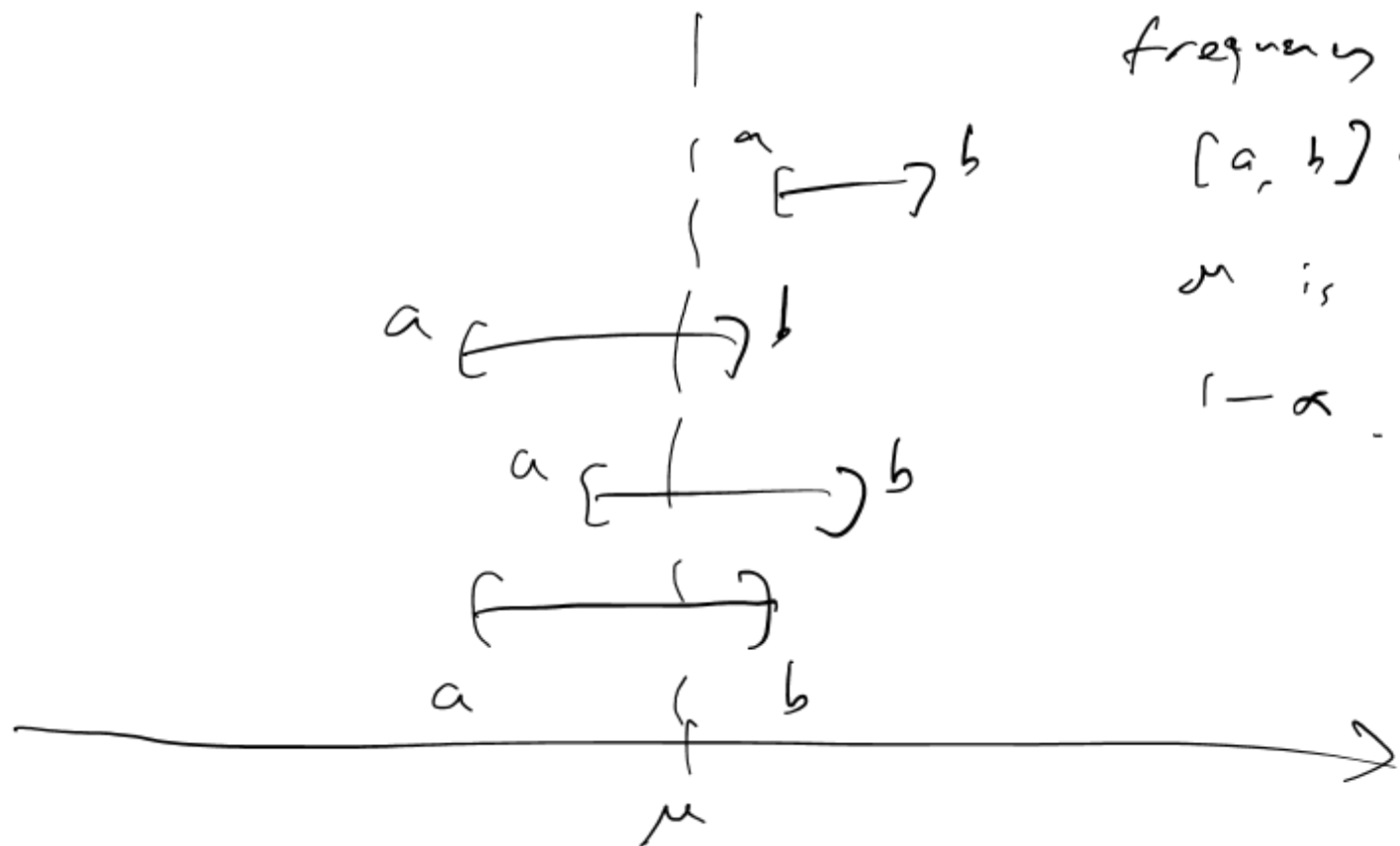
- Suppose $X_1, X_2, \dots, X_n$ is an i.i.d. sample from a population with an unknown parameter, say μ
- To quantify error, we need an interval estimate, or confidence interval: Find two numbers, say a and b , from the simulation replications or data, such that the interval $[a, b]$ contains μ with high confidence, e.g.,

$$P(a \leq \mu \leq b) = 1 - \alpha$$

in this prob,
 a & b are
random because
they depend
on data

- This is called the $(1 - \alpha)$ -level *confidence interval*
- Typically $1 - \alpha = 95\%, 90\%, 99\%$ etc.

μ is deterministic
but unknown



frequency of
 $[a, b]$ covering
 μ is
 $1 - \alpha$.

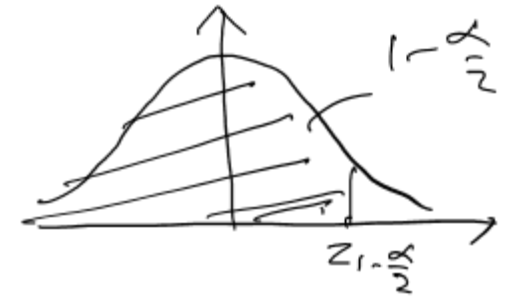
Confidence Interval: Known σ

- Suppose $X_1, X_2, \dots, X_n$ is a sample from a population with unknown mean μ and known variance σ^2
- The $(1 - \alpha)$ confidence interval is

$$\left[\bar{X} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right]$$

where $z_{1-\frac{\alpha}{2}}$ is called the z-critical value = $\left(1 - \frac{\alpha}{2}\right)$ -quantile of $N(0,1)$, i.e.,

$$P(Z \leq z_{1-\frac{\alpha}{2}}) = 1 - \frac{\alpha}{2}$$



E.g.,

- for 95% confidence interval, $\alpha = 5\%$, and we use $z_{0.975} = 1.96$
- For 90% confidence interval, $\alpha = 10\%$ and we use $z_{0.95} = 1.645$

Do we need X_i to follow a normal distribution?

Explanation of the CI formula:

$$\bar{X} = \frac{x_1 + \dots + x_n}{n}$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

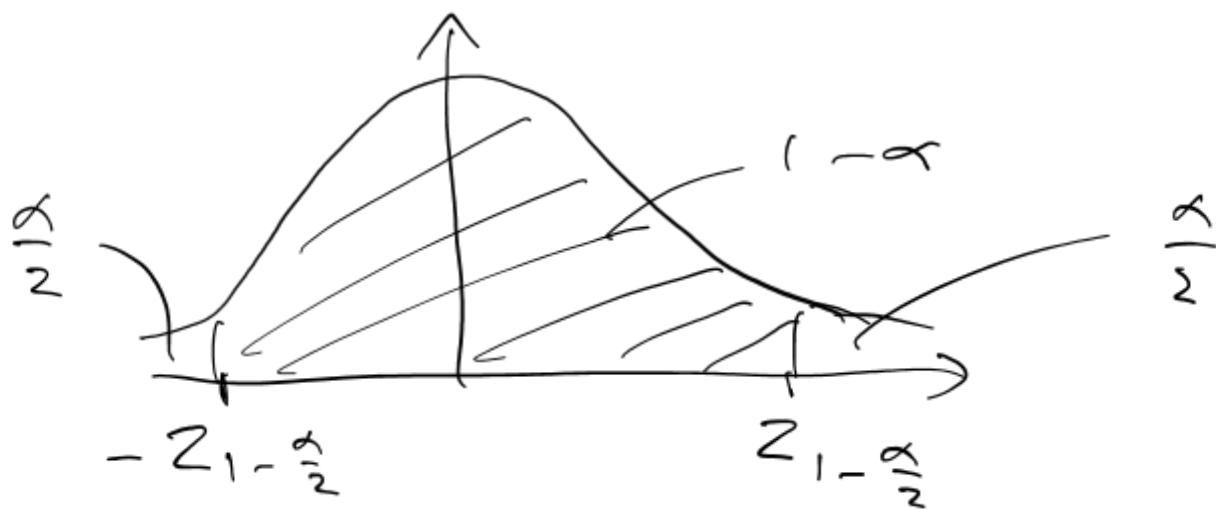
approximately when n is large
(by CLT)

or when $x_1, \dots, x_n$ are normal.

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

$$P(-z_{1-\frac{\alpha}{2}} \leq Z \leq z_{1-\frac{\alpha}{2}}) = 1 - \alpha$$

where
 $Z \sim N(0, 1)$



So

$$P\left(-z_{1-\frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left(-z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$P\left(\underbrace{\bar{X} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}}_a \leq \mu \leq \underbrace{\bar{X} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}}_b\right) = 1 - \alpha$$

Requirement for the CI to work:

n is large enough ($n \geq 30$, s.t. that)
CLT holds

or data are normally distributed

Confidence Interval: Known σ

Explanation:

- If X_i are normal or when CLT is applicable, we have

$$P\left(-x \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq x\right) \approx P(-x \leq Z \leq x)$$

where $Z \sim N(0,1)$

- Note that

$$P(-z_{0.975} \leq Z \leq z_{0.975}) = \Phi(z_{0.975}) - \Phi(-z_{0.975}) = 0.95$$

- So

$$P\left(-z_{0.975} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{0.975}\right) \approx 0.95$$

- Rearranging terms, this becomes

$$P\left(\bar{X} - z_{0.975} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{0.975} \frac{\sigma}{\sqrt{n}}\right) \approx 0.95$$

Confidence Interval: Known σ

When does this interval hold?

- When X_i are normal, the *coverage* holds exactly correct, for any n , i.e.,

$$P\left(\bar{X} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

- When n is large enough ($n \geq 30$), CLT applies and the coverage holds approximately correct, i.e.,

$$P\left(\bar{X} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) \approx 1 - \alpha$$

Confidence Interval: Known σ

- **Example.** A manager at a beverage company knows that the beverage volume inside a can follows a normal distribution with standard deviation $\sigma = 0.2$ ounce. From a sample of 100 cans he obtains the sample mean to be 12.09 ounce. What is a 95% confidence interval for the population mean?

$$\left[\bar{x} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \quad \bar{x} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right]$$

$$= \left[12.09 - 1.96 \frac{0.2}{\sqrt{100}}, \quad 12.09 + 1.96 \frac{0.2}{\sqrt{100}} \right]$$

$$= [11.99, 12.19]$$

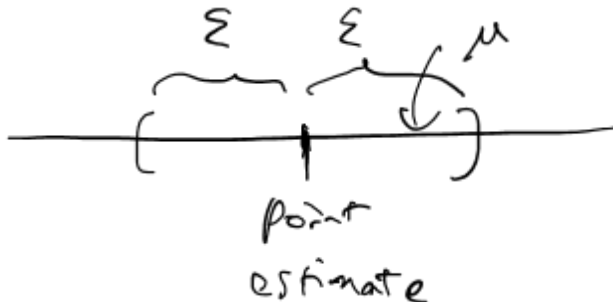
Confidence Interval: Known σ

- **Example.** A manager at a beverage company knows that the beverage volume inside a can follows a normal distribution with standard deviation $\sigma = 0.2$ ounce. From a sample of 100 cans he obtains the sample mean to be 12.09 ounce. What is a 95% confidence interval for the population mean?
- **Solution.** Plugging formulas from previous slide, the interval takes the form

$$\left[12.09 - 1.96 \times \frac{0.2}{\sqrt{100}}, 12.09 + 1.96 \times \frac{0.2}{\sqrt{100}} \right]$$
$$= [12.09 - 0.039, 12.09 + 0.039] = [12.05, 12.13]$$

Sample Size Determination: Known σ

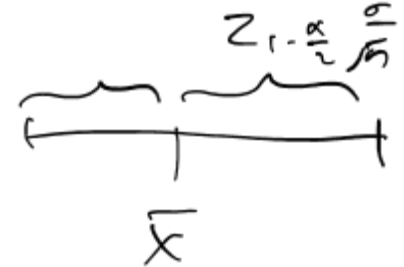
- Sometimes, before full data collection, we want to know how much data is needed
- We want a sample size n to guarantee:
 - A (point) estimate is within an error ϵ from the truth with $1 - \alpha$ probability
 - The half-width of the $1 - \alpha$ confidence interval is within ϵ
 - The margin of error at the $1 - \alpha$ confidence level is within ϵ
- These criteria are all equivalent



Sample Size Determination: Known σ

We need a sample size n such that

$$z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \epsilon$$



or that

$$n \geq \left(z_{1-\frac{\alpha}{2}} \frac{\sigma}{\epsilon} \right)^2$$

We round up the required sample size n to an integer

Note: This procedure is valid only when

- The population is normal or
- The obtained n at the end is sufficiently large, i.e., ≥ 30

Sample Size Determination: Known σ

- **Example.** From past experience it is known that the weights of salmon grown at a commercial hatchery are normal with a mean that varies from season to season but with a standard deviation that remains fixed at 0.3 pounds. If we want to be 95% confident that our estimate of the present season's mean salmon weight is correct within ± 0.1 pounds, how large a sample is needed?

$$\begin{aligned} n &\geq \left(z_{1-\frac{\alpha}{2}} \frac{\sigma}{\bar{\Delta}} \right)^2 \\ &= \left(1.96 \frac{0.3}{0.1} \right)^2 \\ &\quad \dots \text{round up} \end{aligned}$$

The determined sample size can be any number in this example because the population is normal.

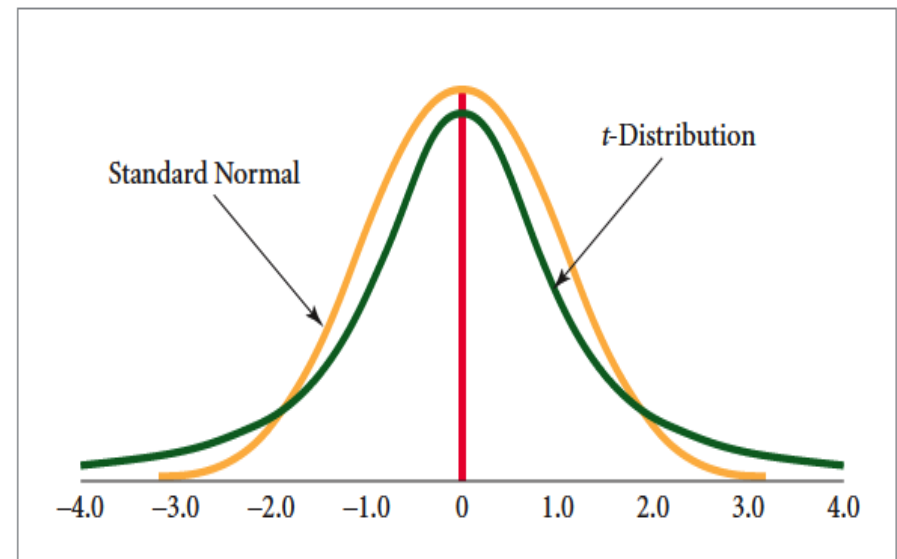
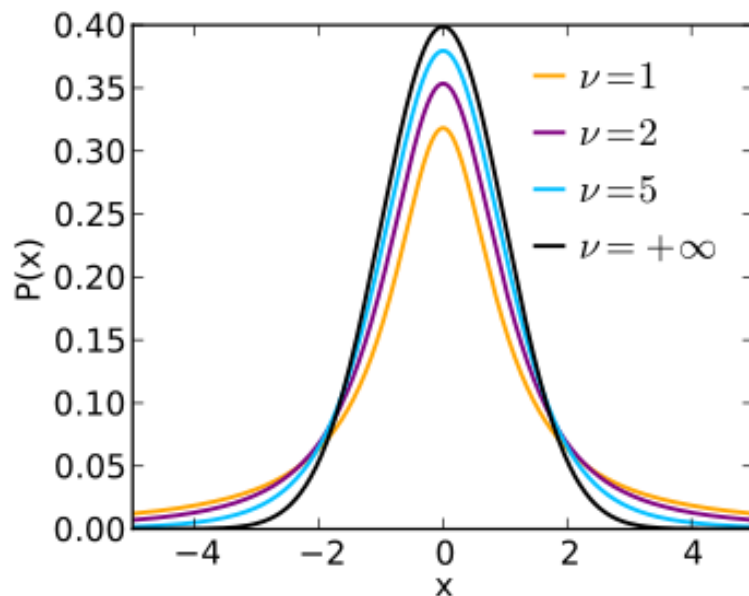
In other cases where the population is not normal, we need to ensure the determined sample size is at least 30.

Sample Size Determination: Known σ

- **Example.** From past experience it is known that the weights of salmon grown at a commercial hatchery are normal with a mean that varies from season to season but with a standard deviation that remains fixed at 0.3 pounds. If we want to be 95% confident that our estimate of the present season's mean salmon weight is correct within ± 0.1 pounds, how large a sample is needed?
- **Solution.** Let the population mean be μ . We want $\frac{1.96\sigma}{\sqrt{n}} = \frac{0.588}{\sqrt{n}} \leq 0.1$ or $n \geq \left(1.96 \frac{\sigma}{0.1}\right)^2$ giving $n \geq 34.57$. The required sample size is 35.

Confidence Interval: Unknown σ

- A digression: t-distribution with $(n-1)$ degrees of freedom
- Like $N(0,1)$, t-distribution is symmetric around 0
- t-distribution has a “heavier” tail than normal
- t-distribution has lighter tail as the degree of freedom increases, and is very similar to $N(0,1)$ when n is large, e.g., $n \geq 30$



Confidence Interval: Unknown σ

Define

- Sample variance $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- Sample standard deviation $S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$

Confidence Interval: Unknown σ

- Suppose $X_1, X_2, \dots, X_n$ is a sample from a population with unknown mean μ and unknown variance σ^2
- If n is small, i.e., < 30 , and X_i is normally distributed, then we use

$$\left[\bar{X} - t_{1-\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{1-\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \right]$$

as the $(1 - \alpha)$ confidence interval for μ

- Here $t_{1-\frac{\alpha}{2}, n-1}$ is called the t -critical value = $\left(1 - \frac{\alpha}{2}\right)$ -quantile of a t -distribution with degree of freedom $n - 1$, i.e.,

$$P(T_{n-1} \leq t_{1-\frac{\alpha}{2}, n-1}) = 1 - \frac{\alpha}{2}$$

where T_{n-1} is a t -variable with degree of freedom $n - 1$

Confidence Interval: Unknown σ

Justification:

Let $X_1, X_2, \dots, X_n$ be a sample from a normal population with mean μ and variance σ^2 . Define the sample mean $\bar{X}$ and sample variance S^2 . Then, $\sqrt{n}(\bar{X} - \mu)/S$ has a Student's t -distribution with $(n-1)$ degrees of freedom

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1} \quad \text{if } X_1, \dots, X_n \text{ are normally distributed}$$

Confidence Interval: Unknown σ

- Suppose $X_1, X_2, \dots, X_n$ is a sample from a population with unknown mean μ and unknown variance σ^2

- If n is large, i.e., ≥ 30 , then we use

$$\left[\bar{X} - z_{1-\frac{\alpha}{2}} \frac{S}{\sqrt{n}}, \bar{X} + z_{1-\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \right]$$

as the $(1 - \alpha)$ confidence interval for μ

- Justified from the CLT and a plug-in of the sample variance to approximate the unknown variance

Confidence Interval: Unknown σ

Note on notation:

- In the textbook, z_α and $t_{\alpha,n}$ are the $(1 - \alpha)$ -quantile of $N(0,1)$ and T_n respectively
- They refer to the lower quantile, whereas the slides refer to the upper quantiles
- Both notations are common

TABLE A3 Values of $t_{\alpha,n}$

n	$\alpha = .10$	$\alpha = .05$	$\alpha = .025$	$\alpha = .01$	$\alpha = .005$
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.474	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
∞	1.282	1.645	1.960	2.326	2.576

Confidence Interval: Unknown σ

- **Example.** A regional certified public accounting (CPA) firm near Minneapolis conducted an audit for a discount chain. One part of the audit involved developing an estimate for the mean dollar error in total charges that occurs during checkout. The firm wishes to develop a 90% CI estimate for the population mean using a sample of size 20 with the following data (positive values indicate overcharge):

\$0.00	\$1.20	\$0.43	\$1.00	\$1.47	\$0.83	\$0.50	\$3.34	\$1.58	\$1.46
-\$0.36	-\$1.10	\$2.60	\$0.00	\$0.00	-\$1.70	\$0.83	\$1.99	\$0.00	\$1.34

- Assume a dollar error is normally distributed. Determine a 90% CI for the mean dollar error

Confidence Interval: Unknown σ

- **Answer:**

- $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = 0.7705$

- $S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2} = 1.1936$

- 90% confidence interval

$$= \left[\bar{X} - t_{.95,19} \frac{S}{\sqrt{n}}, \bar{X} + t_{.95,19} \frac{S}{\sqrt{n}} \right]$$

$$= \left[0.7705 - 1.729 \times \frac{1.1936}{\sqrt{20}}, 0.7705 + 1.729 \times \frac{1.1936}{\sqrt{20}} \right]$$

$$= [0.31, 1.23]$$

Sample Size Determination: Unknown σ

To guarantee:

- A (point) estimate is within an error ϵ from the truth with $1 - \alpha$ probability
- The half-width of the $1 - \alpha$ confidence interval is within ϵ
- The margin of error at the $1 - \alpha$ confidence level is within ϵ

we need a sample size n such that

$$z_{1-\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \leq \epsilon$$

or that

$$n \geq \left(z_{1-\frac{\alpha}{2}} \frac{S}{\epsilon} \right)^2$$

We round up the required sample size n to an integer

Sample Size Determination: Unknown σ

Notes:

- The sample variance S^2 needs to be estimated from a “pilot” set of data
- This procedure is valid only when the obtained n at the end is sufficiently large, i.e., ≥ 30
- When the obtained $n < 30$, then using the z -critical value was no longer valid; instead one should switch to using t -critical value

Confidence Interval: Unknown σ

Continuing and given the information from the example, how many more data points are needed to attain a margin of error in estimating the mean dollar error to within ± 0.01 with 90% confidence?

Confidence Interval: Unknown σ

Continuing and given the information from the example, how many more data points are needed to attain a margin of error in estimating the mean dollar error to within ± 0.01 with 90% confidence?

Ans:

$$\bullet \quad n \geq \left(z_{1-\frac{\alpha}{2}} \frac{s}{\epsilon} \right)^2 = \left(1.645 \times \frac{1.1936}{0.01} \right)^2 = 38552.22$$

Number of additional data points needed = $38553 - 20 = 38533$

Confidence Interval: Proportion

- Suppose the data are Bernoulli, i.e., each individual outcome is either a “success” (1) or “failure” (0).
- We are interested in estimating p , the proportion of success
- We can look at the sample proportion as the point estimate

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$$

where each data point X_i is either 0 or 1.

Confidence Interval: Proportion

- We also know that the variance of X_i is $p(1 - p)$, which can be approximated by $\hat{p}(1 - \hat{p})$
- Therefore, the $(1 - \alpha)$ confidence interval for p is

$$\left[\hat{p} - z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}, \hat{p} + z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \right]$$

- This is valid if n is a sufficiently large number, i.e., ≥ 30 so that a central limit theorem holds

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \sim N(0,1) \text{ approximately}$$

Sample Size Determination: Proportion

To guarantee:

- A (point) estimate is within an error ϵ from the truth with $1 - \alpha$ probability
- The half-width of the $1 - \alpha$ confidence interval is within ϵ
- The margin of error at the $1 - \alpha$ confidence level is within ϵ

we need a sample size n such that

$$z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq \epsilon$$

or that

$$n \geq z_{1-\frac{\alpha}{2}}^2 \frac{\hat{p}(1-\hat{p})}{\epsilon^2}$$

We round up the required sample size n to an integer

Notes:

- The sample variance $\hat{p}$ is estimated from a “pilot” set of data; otherwise we simply take $\hat{p} = 1/2$ as this is the most conservative
- This procedure is valid only when the obtained n at the end is sufficiently large, i.e., ≥ 30

Example

A sample of 100 transistors is randomly chosen from a large batch and tested to determine if they meet the current standards. If 80 of them meet the standards, what is an approximate 95 percent confidence interval for the fraction of all the transistors that meet the standards?

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Ans:

$$\hat{p} = \frac{80}{100} = 0.8$$

95% confidence interval

$$= \left[0.8 - 1.96 \sqrt{\frac{0.8 \times (1-0.8)}{100}}, 0.8 + 1.96 \sqrt{\frac{0.8 \times (1-0.8)}{100}} \right] = [0.72, 0.88]$$

Example

Continuing and given the information in the previous example, how many more transistors should be sampled from in order to get a margin of error in estimating the fraction that meets the standard within $\pm 1\%$ with 95% confidence?

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Ans:

$$n \geq 1.96^2 \times \frac{0.8(1 - 0.8)}{0.01^2} = 6146.56$$

Number of additional transistors to be sampled
 $= 6147 - 100 = 6047$

Example

Continuing in the previous example, but now assume that you don't have any information about the transistors to begin with. How many transistors should be sampled from in order to get a margin of error in estimating the fraction that meets the standard within $\pm 1\%$ with 95% confidence?

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Continuing in the previous example, but now assume that you don't have any information about the transistors to begin with. How many transistors should be sampled from in order to get a margin of error in estimating the fraction that meets the standard within $\pm 1\%$ with 95% confidence?

Ans:

$$n \geq 1.96^2 \times \frac{0.5(1 - 0.5)}{0.01^2} = 9604$$

Number of transistors to be sampled = 9604